

SESAN D

AI Systems Engineer | Computer Vision & Edge AI

Coimbatore, Tamil Nadu, India

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PROFESSIONAL SUMMARY

AI Systems Engineer with 2 years of hands on experience building and deploying real time computer vision and intelligent automation systems. Specialized in GPU optimized inference, multi camera video analytics, edge deployment, and scalable FastAPI based AI services. Experienced in integrating LLM and agent driven workflows into production systems across cloud and on premise environments.

WORK EXPERIENCE

AI Systems Engineer

PrimeMover Solutions, Coimbatore

May 2024 – Present

- Designed and deployed real time computer vision systems for surveillance, healthcare, and identity management use cases.
- Built production ready inference pipelines using Python, FastAPI, OpenCV, and PyTorch with GPU optimized deployment.
- Developed RTSP based multi camera streaming architectures with stable real time performance.
- Implemented face recognition, ANPR, object detection, and anti spoof validation systems.
- Optimized inference latency using NVIDIA GPU acceleration and ONNX Runtime based model optimization.
- Deployed scalable AI services using Docker across cloud and on premise GPU environments.
- Integrated LLM driven automation workflows and agent based API orchestration into production systems.

PROJECTS

SnapFace – Real Time Face Recognition System

- Built dual RTSP based face recognition system with anti spoof validation and automated alerts.
- Implemented GPU optimized InsightFace inference using ONNX Runtime for stable real time performance.
- Designed secure role based access and live monitoring dashboard.
- Handled multi camera streaming with consistent performance under continuous video input.

Tech: Python, FastAPI, InsightFace, OpenCV, ONNX Runtime

SpeedVision AI – Video Analytics & ANPR System

- Developed real time video analytics pipeline for detection, tracking, ANPR, and speed estimation.
- Built RTSP streaming pipeline with GPU accelerated YOLOv8 inference.
- Integrated OCR based license plate recognition and structured alert logging backend.
- Optimized low latency inference for continuous surveillance monitoring.

Tech: YOLOv8, OpenCV, FFmpeg, Tesseract, FastAPI

MedToolVision – Edge Surgical Tool Tracking

- Designed edge deployed surgical instrument tracking system on Raspberry Pi CM4.
- Implemented low latency detection pipeline with HDMI live visualization output.
- Optimized inference performance for constrained edge hardware environments.

Tech: Python, OpenCV, YOLOv8

CORE EXPERTISE

- Real time video analytics systems
 - Edge AI deployment and GPU accelerated inference
 - Multi camera streaming architecture and RTSP pipelines
 - Face recognition and anti spoof validation
 - LLM based automation and RAG workflows
 - Production ready FastAPI services
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TECHNICAL SKILLS

- Languages: Python, SQL
 - Computer Vision and ML: OpenCV, NumPy, PyTorch, TensorFlow; YOLOv8, InsightFace; object detection, tracking, OCR
 - LLM and Agents: RAG pipelines, tool calling, LangChain, LangGraph, CrewAI
 - Backend and APIs: FastAPI, REST APIs, WebSockets
 - Streaming and Optimization: FFmpeg, RTSP pipelines, ONNX Runtime, OpenVINO, FP16 and INT8 quantization
 - Deployment and Cloud: Docker, Podman, Microsoft Azure, on premise GPU servers
 - Data and Annotation: CVAT, Roboflow
 - Developer Tools and OS: GitHub, VS Code, Linux, Windows
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INTERNSHIP EXPERIENCE

Machine Learning Engineer Intern

Center for Machine Learning and Intelligence, Coimbatore

December 2023 – April 2024

Developed a Raspberry Pi based postal sign recognition system using computer vision and machine learning models. Implemented image preprocessing, model training, and lightweight edge inference for real time detection on embedded hardware.

EDUCATION

Master of Computer Applications (MCA)

Gobi Arts & Science College, Tamil Nadu

2022 – 2024 | CGPA: 8.54

Bachelor of Science in Mathematics

Gobi Arts & Science College, Tamil Nadu

2019 – 2022 | CGPA: 8.08

CERTIFICATIONS & ACHIEVEMENTS

- Ideathon 24 Winner – PrimeMover Solutions
 - Machine Learning Internship Certificate – Avinashilingam University
 - AI for IoT Speech Appreciation Letter – Gobi Arts and Science College
 - Employee Upliftment Program Certificate – Skill Hub
 - Python Programming Certification – Udemy
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LANGUAGES

- Tamil - Native
 - English - Professional Working Proficiency
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